

- 2) Determine if the following limit exist or not. If it exists, give the value of the limit.

$$\lim_{(x,y) \rightarrow (1, -1)} (x^2 + 2xy + y^2) / (x^3 + y^3),$$

The function is not continuous at point $(1, -1)$, as at this point, the denominator becomes zero. However, **except for** the line $x + y = 0$, at which the point $(1, -1)$ lies and at which the limit is intended to be calculated, *the function* $(x^2 + 2xy + y^2) / (x^3 + y^3)$ equals the function $(x + y) / (x^2 - xy + y^2)$. And, we know, in calculating the limit at a point (here $(x, y) = (1, -1)$), the value at the point is NOT taken into consideration. Therefore, for the purpose of limit

$$\begin{aligned} \lim_{(x,y) \rightarrow (1, -1)} [(x^2 + 2xy + y^2) / (x^3 + y^3)] &= \lim_{(x,y) \rightarrow (1, -1)} [(x + y) / (x^2 - xy + y^2)] \\ &= \lim_{(x,y) \rightarrow (1, -1)} [(1 - 1) / (1^2 - (1)(-1) + (-1)^2)] = 0/3 = 0 \end{aligned}$$

The limit has been calculated by removing the problematic condition, i. e. *division by zero*, at the point/line where limit is to be computed.

- 3) Determine if the following limit exist or not. If it exists, give the value of the limit.

$$\lim_{(x,y) \rightarrow (1,1)} [2(1-y) + 3(x-1)] / (x-y)$$

In this case, **(i)** function is not continuous at the point $(1, 1)$, as the denominator becomes 0 at $(1, 1)$ and **(ii)** no factor which makes denominator 0, can be cancelled out. **Therefore, there is possibility of the limit of the function NOT existing at point $(1, 1)$.** If the limit, actually, does not exist, and to show so, we need to find two paths to the point $(1, 1)$ along which the proposed limits differ.

A path on a surface of a function of two variables is obtained generally either (i) by taking one of the variables as constant; or (ii) replacing one of the variables as function of the other, e. g. $y=x$, or $y = 2x+1$ etc.

We try some paths, which may lead to differing values of limit, and hence establishing non-existence of limit.

First, **we take the path $x=1$** , for which

$$[2(1-y) + 3(x-1)] / (x-y) \text{ becomes } [2(1-y)] / (1-y) = 2$$

Therefore,

$$\lim_{(1,y) \rightarrow (1,1)} [2(1-y) + 3(x-1)] / (x-y) = \lim_{(1,y) \rightarrow (1,1)} 2 = 2$$

Then, **we take the path $y=1$** , for which

$$[2(1-y) + 3(x-1)] / (x-y) \text{ becomes } 3((x-1)) / (x-1) = 3$$

Therefore

$$\lim_{(x,1) \rightarrow (1,1)} [2(1-y) + 3(x-1)] / (x-y) = \lim_{(x,1) \rightarrow (1,1)} 3 = 3$$

The two proposed limits along two different paths do not coincide, hence the required limit does not exist.

Check Your Progress 2

1) $z = 8x / (x^2 + 5y)$

$$\partial z / \partial x = \{8(x^2 + 5y) - 8x(2x)\} / (x^2 + 5y)^2; \text{ treating } y \text{ as a constant}$$

$$= \{8x^2 + 40y - 16x^2\} / (x^2 + 5y)^2$$

$$\partial z / \partial y = 8x \{(-1)(x^2 + 5y)^{-2}(5)\}; \text{ treating } x \text{ as a constant}$$

$$= -40x / (x^2 + 5y)^2$$

2) $f(x, y, z) = x \sin(y) / z^2$

$$\partial f / \partial x = \sin(y) / z^2; \text{ treating } y \text{ and } z \text{ as constants}$$

$$\partial f / \partial y = x \cos(y) / z^2; \text{ treating } x \text{ and } z \text{ as constants}$$

$$\partial f / \partial z = x \sin(y) (-2) / z^3; \text{ treating } y \text{ and } z \text{ as constants}$$

3) $z = \sqrt{(x^2 + \log(5x - 3y^2))}$

$$\partial z / \partial x = \{1/2 / ((x^2 + \log(5x - 3y^2))^{1/2})\} \{2x + (5/(5x - 3y^2))\}$$

$$= \dots \dots$$

$$\partial z / \partial y = \{1/2 / ((x^2 + \log(5x - 3y^2))^{1/2})\} \{-6y/(5x - 3y^2)\}$$

$$= (-3y / (5x - 3y^2)) (x^2 + \log(5x - 3y^2))^{-1/2} = \dots$$

4) Find $\partial^3 f / \partial y \partial x^2$, for $f(x, y) = e^{xy}$

By definition

$$\partial^3 f / \partial y \partial x^2 = \partial / \partial y (\partial / \partial x (\partial / \partial x (e^{xy})))$$

$$= \partial / \partial y (\partial / \partial x \{y e^{xy}\}); \text{ taking } y \text{ as a constant}$$

$$= \partial / \partial y \{y (y e^{xy})\} = \partial / \partial y \{y^2 e^{xy}\}; \text{ taking } y \text{ as a constant}$$

$$= (e^{xy}) \partial / \partial y (y^2) + (y^2) \partial / \partial y (e^{xy}); \text{ taking } x \text{ as a constant}$$

$$= 2y (e^{xy}) + (y^2)(x e^{xy})$$

$$= (2y + xy^2) (e^{xy})$$

5) For the function $z = f(x, y) = 5x^3 y^4$, calculate $z_{xxx}, z_{xxy}, z_{xyx}, z_{xyy}, z_{yyy}$

From section 14.4.4, we know that

$$z_{xx} = f_{xx} = (f_x)_x = \partial^2 f / \partial x^2 = 30y^4 x.$$

$$z_{xy} = f_{xy} = (f_x)_y = \partial / \partial y (\partial f / \partial x) = \partial^2 f / \partial y \partial x = 60y^3 x^2$$

Therefore

$$z_{xxx} = f_{xxx} = (f_{xx})_x = \partial^3 f / \partial x^3 = \partial / \partial x (\partial^2 f / \partial x^2) = \partial / \partial x (30y^4 x) = 30y^4$$

$$z_{xxy} = f_{xxy} = (f_{xx})_y = \partial^3 f / \partial y \partial x^2 = \partial / \partial y (\partial^2 f / \partial x^2) = \partial / \partial y (30y^4 x) = 120y^3 x$$

$$z_{xyx} = (f_{xy})_x = \partial^3 f / \partial x \partial y \partial x = \partial / \partial x (f_{xy}) = \partial / \partial x (60y^3x^2) = 120 y^3 x$$

$$z_{xyy} = (f_{xy})_y = \partial^3 f / \partial x \partial y \partial y = \partial / \partial y (f_{xy}) = \partial / \partial y (60y^3x^2) = 180 y^2 x^2$$

For z_{yyy} , we know

$$z = f(x, y) = 5x^3 y^4$$

$$z_y = f_y(x, y) = 20 x^3 y^3$$

$$z_{yy} = f_{yy} = (\partial^2 f / \partial y^2) = 60 x^3 y^2$$

$$z_{yyy} = f_{yyy} = (f_{yy})_y = \partial^3 f / \partial y^3 = \partial / \partial y (\partial^2 f / \partial y^2) = 120 x^3 y$$

- 6) The vector $\vec{u}$ is specified by $\langle \cos \pi/4, \sin \pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2} \rangle$

Thus, directional derivative is given as

$$D\vec{u}(x, y) = 1/\sqrt{2} f_x + 1/\sqrt{2} f_y = 1/\sqrt{2} \{f_x + f_y\} = \dots$$

$$D\vec{u}(x, y) = (1/\sqrt{2}) [\{2x \tan y + 3 \log y\} + \{x^2 \sec^2 y + 3(x/y)\}]$$

$$D\vec{u}(1, \pi/3) = (1/\sqrt{2}) [2 \tan \pi/3 + 3 \log(\pi/3) + \sec^2(\pi/3) + 9/\pi] = \dots$$

Check Your Progress 3

- 1) Let $f_1(x, y) = \sin x$, $f_2(x, y) = 2xy$, $f_3(x, y) = 3xy^2$.

$$J_f = \begin{bmatrix} \partial f_1 / \partial x & \partial f_1 / \partial y \\ \partial f_2 / \partial x & \partial f_2 / \partial y \\ \partial f_3 / \partial x & \partial f_3 / \partial y \end{bmatrix} = \begin{bmatrix} \cos x & 0 \\ 2y & 2x \\ 3y^2 & 6xy \end{bmatrix}$$

Therefore, $(Jf)_{(\pi, \pi/2)} = \pi \begin{bmatrix} -1 & 0 \\ 2\pi & 2\pi \\ 3\pi^2/4 & 3\pi^2 \end{bmatrix}$

2)

$$\text{Let } f_1(x, y, z) = 2z + \sin x, \quad f_2(x, y, z) = 2xy - z^2, \quad f_3(x, y, z) = 3xy^2 - 3z.$$

$$J_f = \begin{bmatrix} \partial f_1 / \partial x & \partial f_1 / \partial y & \partial f_1 / \partial z \\ \partial f_2 / \partial x & \partial f_2 / \partial y & \partial f_2 / \partial z \\ \partial f_3 / \partial x & \partial f_3 / \partial y & \partial f_3 / \partial z \end{bmatrix} = \begin{bmatrix} \cos x & 0 & 2 \\ 2y & 2x & -2z \\ 3y^2 & 6xy & -3 \end{bmatrix}$$

Therefore, $(Jf)_{(\pi, \pi/2, 1)} = \begin{bmatrix} -1 & 0 & 2 \\ \pi & 2\pi - 2 & -2 \\ 3\pi^2/4 & 3\pi^2 - 3 & -3 \end{bmatrix}$

- 3) First, we calculate first order partial derivatives

$$\begin{bmatrix} \partial f / \partial x & \partial f / \partial y & \partial f / \partial z \end{bmatrix} = \begin{bmatrix} 6y^2 & 6x & 6y^2 \end{bmatrix}$$

Therefore

$$H_f = \begin{bmatrix} \partial^2 f / \partial x^2 & \partial^2 f / \partial x \partial y \\ \partial^2 f / \partial y \partial x & \partial^2 f / \partial y^2 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 12y \end{bmatrix}$$

where

$\partial^2 f / \partial x^2$ etc denote second order partial derivative of f etc. w. r. t. x .

$$(H_f)_{(1,2)} = \begin{bmatrix} 6 & 0 \\ 0 & 24 \end{bmatrix}$$

Check Your Progress 4

- 1) We know from our earlier knowledge of calculus that *every polynomial* is differentiable on its domain of definition. Hence, both the conditions of Mean value theorem are satisfied. Therefore, there exists $c \in]-1, 2[$ s. t.

$$f'(c) = \{f(2) - f(-1)\} / (2 - (-1)) = \{f(2) - f(-1)\} / (3)$$

Differentiating the given function

$$f(y) = y^3 + 2y^2 - y$$

w.r.t y , we get

$$f'(y) = 3y^2 + 4y - 1$$

By mean value theorem, c satisfies

$$3c^2 + 4c - 1 = [f(2) - f(-1)] / \{2 - (-1)\} = (14 - 2) / 3 = 4$$

$3c^2 + 4c - 5 = 0$, which is a quadratic equation. Solving the quadratic equation for c we get two values for c as

$$c = (-4 \pm \sqrt{76}) / 6 =$$

i.e.

$$c_1 = (-2 + \sqrt{19}) / 3 = 0.7863 \quad \text{and} \quad c_2 = (-2 - \sqrt{19}) / 3 = -2.1196$$

However, c_2 is not in the interval $]-1, 2[$.

Hence, $c_1 = 0.7863$ is the only required value.

- 2) Since f is a polynomial, f is continuous on the closed interval $[0, 1/2]$ and derivable on the open interval $]0, 1/2[$. Hence, by mean value theorem, there exists $c \in]0, 1/2[$ such that

$$f'(c) = (f(1/2) - f(0)) / (1/2 - 0) \dots \dots (i)$$

For knowing the value of $f'(c)$, the derivative of $f(x)$ is given by

$$f'(x) = (x-1)(x-2) + x(x-1) + x(x-2)$$

$$\text{or } f'(x) = 3x^2 - 6x + 2. \dots \dots \dots \text{(ii)}$$

$$\text{Also, } f(1/2) = 3/8, \text{ and } f(0) = 0 \dots \dots \text{(iii)}$$

From (i), (ii) and (iii), we get

$$3c^2 - 6c + 2 = (3/8) / (1/2) = 3/4, \text{ implying}$$

$$12c^2 - 24c + 5 = 0. \quad \text{Hence, } c = (6 \pm \sqrt{21}) / 6$$

Out of these two values of c only $1 - (\sqrt{21} / 6)$ lies in $]0, 1/2[$. Hence, the required value is $1 - (\sqrt{21} / 6)$ and the theorem is verified.

- 3) We calculate values at $x = 0$ of $\cos(x)$ and of its various higher order derivatives, starting with the first derivative.

$$f(x)_{x=0} = \cos(x)_{x=0} = 1; f'(x)_{x=0} = -\sin(x)_{x=0} = 0; f''(x)_{x=0} = -\cos(x)_{x=0} = -1,$$

$$f'''(x)_{x=0} = \sin(x)_{x=0} = 0; \quad \dots \dots \dots (\text{%%3})$$

Part (i)

Linear approximation of $f(x) = P_{1,0} = f(0) + x (f'(0))$, where the remainder term $R_1(x)$ is neglected

$$\text{Linear approximation of } \cos(x) = \cos(0) + x (\sin(x))_{x=0} = 1 + 0 = 1.$$

Quadratics approximation of $f(x) = P_{2,0} = f(0) + x f'(x) + (x^2 / 2!) f''(x)$, where the remainder term $R_2(x)$ is neglected.

$$\begin{aligned} \text{Quadratics approximation of } \cos(x) &= P_{2,0} = (\cos x)_{x=0} + x (f'(\cos x))_{x=0} \\ &+ (x^2 / 2!) (f''(\cos x))_{x=0} \\ &= 1 - (x^2 / 2!). \end{aligned}$$

Part (ii) Maclaurin's expansion

The sequence of values of the derivatives in (%%3) above repeats as follows:

$$f^{iv}(\cos(x))_{x=0} = 1; \quad f'(\cos(x))_{x=0} = 0; \quad f^{vi}(\cos(x))_{x=0} = -1; \quad f^{vii}(\cos(x))_{x=0} = 0.$$

The repeating of the above sequence of the values of derivatives may be further generalized as follows:

$$f^{4k}(\cos(x))_{x=0} = 1; \quad f^{4k+1}(\cos(x))_{x=0} = 0; \quad f^{4k+2}(\cos(x))_{x=0} = -1; \quad f^{4k+3}(\cos(x))_{x=0} = 0.$$

$$\text{for } k = 0, 1, 2, 3, 4, \dots$$

Substituting these values in Maclaurin's series expansion given by

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \dots + \frac{f^{(n)}(0)}{n!}x^n + \dots$$

we get: $\cos(x) = 1 + x \cdot 0 + (x^2/2!) \cdot (-1) + (x^3/3!) \cdot (0) + (x^4/4!) \cdot 1 + \dots$
 $= 1 - (x^2/2!) + (x^4/4!) - (x^6/6!) + (x^8/8!) + \dots$

- 4) For the function $f(x) = \log x$, around the point $x=1$, find (i) linear and quadratic approximations, and (ii) Taylor's series expansion of the function.

(Note: $\log(x)$ is NOT defined at $x=0$. Hence, Maclaurin's series is NOT defined)

We calculate values at $x=1$, of $f(x) = \log(x)$ and of its various order derivatives, starting with the first derivative.

$$f(x)_{x=1} = \log(x)_{x=1} = 0;$$

$$f'(x)_{x=1} = (1/x)_{x=1} = 1;$$

$$f''(x)_{x=1} = -(1/x^2)_{x=1} = -1,$$

$$f'''(x)_{x=1} = (2/x^3)_{x=1} = 2; \quad \dots \dots \dots (2/x^3)$$

Part (i)

Linear approximation of $f(x) = P_{1,1} = f(1) + (x-1)f'(1)$, where the remainder term $R_1(x)$ is neglected

Linear approximation, around $x=1$, of $\log(x) = P_{1,1} = 0 + (x-1) = (x-1)$.

Quadratics approximation of $f(x) = P_{2,1} = f(1) + (x-1)f'(1) + ((x-1)^2/2!)f''(1)$, where the remainder term $R_2(x)$ is neglected

Quadratics approximation, around $x=1$, of $\log(x) = 0 + (x-1) + ((x-1)^2/2!)(-1) = (x-1) - (x-1)^2/2$

For Taylor's series expansion around $x=1$, of $\log(x)$,

first, we generalize $(2/x^3)$, derivatives of $f(x) = \log x$ and their values at $x=1$.

We obtained

$$f'''(x)_{x=1} = (2/x^3)_{x=1} = 2;$$

In general,

$$f^k(\log x) = [(-1)^{k-1} (k-1)!] / x^k; \text{ Hence,}$$

$$(f^k(\log x))_{x=1} = [(-1)^{k-1} (k-1)!] \dots \dots \dots (2/x^3)$$

Using in Taylor's series expansion, with $a=1$, given in general by #2.1:

$$f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots, \text{ with}$$

$$(f^k(\log x))_{x=1} = [(-1)^{k-1} (k-1)!] \text{ given by (2.1), and } 0! = 1, \text{ we get}$$

$$(\log(x))_{x=1} = 0 + (x-1) - \{(x-1)^2/2\} + \{(x-1)^3/3\} - \{(x-1)^4/4\} + \dots + (-1)^{k-1} (x-1)^k / (k-1) + \dots$$

$$= (x-1) - \{(x-1)^2/2\} + \{(x-1)^3/3\} - \{(x-1)^4/4\} + \dots + (-1)^{k-1} (x-1)^k / (k-1) + \dots$$

14.13 EXERCISES

- 1) Determine if the following limit exist or not. If it exists, give the value of the limit.

$$\lim_{(x,y) \rightarrow (0,0)} [x^4 y] / [x^8 + y^2]$$

- 2) For the function $y=f(x) = (\log (\sin (e^x + 7x^2)))$

Calculate d^2y/dx^2

- 3) Suppose that the amount of water in a holding tank at t minutes is given by $V(t) = 5t^2 - 64t + 45$.

Determine each of the following.

- Is the volume of water in the tank increasing or decreasing at $t = 3$ minute?
- Is the volume of water in the tank increasing or decreasing at $t = 7$ minutes?
- Is the volume of water in the tank changing faster at $t = 3$ or $t = 7$ minutes?

- 4) Calculate the differential for each of the following

a) $y = x^3 \sin(5x)$

b) $f(w) = e^{4-3w}$

c) $z = e^{x \sin(y)}$

d) $g(x, y, z) = x^5 y^3 / z^2$

- 5) Calculate the directional derivative $D_{\vec{u}} f(\vec{x})$ for $f(x; y; z) = \sin(yz) + \log(x^2)$ at $(1, 1, \pi)$ in the direction of $\vec{v} = \langle 1, 1, -1 \rangle$

- 6) Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}^4$ be defined as

$$f((x, y)) = (x \log y, \sin x, x^2 y, y^3),$$

find the Jacobian J_f at the point $(\pi/2, \pi/4)$.

- 7) If f is defined and continuous on closed interval $[a, b]$ and is derivable on the open interval $]a, b[$, and if $f'(x) = 0$, for all x in $]a, b[$, then $f(x)$ has a constant value throughout $[a, b]$.
- 8) If f is continuous on closed interval $[a, b]$ and $f'(x) > 0$ in $]a, b[$, then f is strictly increasing in the interval $[a, b]$.

Answers to Exercises

- 1) In this case, (i) function is not continuous at the point $(0, 0)$, as the denominator becomes 0 at $(0, 0)$ and (ii) no factor which makes denominator 0, can be cancelled out. **Therefore, there is possibility of the limit of the function NOT existing at point $(0, 0)$.** If the limit, actually, does not exist, and to show so, we need to find two paths to the point $(0, 0)$ along which the proposed limits differ.

The popular paths $x=0$, and $y=0$ will not yield the intended result, as along these paths, both the limits will be 0.

We try another well-known path $x=y$, along which

$$\begin{aligned}\lim_{(x,y) \rightarrow (0,0)} [x^4 y] / [x^8 + y^2] &= \lim_{(x,y) \rightarrow (0,0)} [x^4 x] / [x^8 + x^2] \\ &= \lim_{(x,y) \rightarrow (0,0)} x^3 / (x^6 + 1) = 0 \quad \dots \dots \dots \text{(i)}\end{aligned}$$

In this case, from the expression of the function, it appears that the path $y = x^4$ may yield the desired inequality.

Along $y = x^4$

$$\begin{aligned}\lim_{(x,y) \rightarrow (0,0)} [x^4 y] / [x^8 + y^2] &= \\ \lim_{(x,y) \rightarrow (0,0)} [x^4 x^4] / [x^8 + x^8] &= \lim_{(x,y) \rightarrow (0,0)} 1/2 = 1/2. \dots \dots \text{(ii)}.\end{aligned}$$

As (i) and (ii) differ, the required limit does NOT exist.

- 2) We know from 14.4.3.1 that

$$dy/dx = (e^x + 14x) \cot(e^x + 7x^2), \text{ therefore}$$

$$\begin{aligned}d^2y/dx^2 &= d/dx ((e^x + 14x) \cot(e^x + 7x^2)) \\ &= (e^x + 14) \cot(e^x + 7x^2) + (e^x + 14x) [-\operatorname{cosec}^2(e^x + 7x^2)] \times (e^x + 14x) \\ &= (e^x + 14) \cot(e^x + 7x^2) + (e^x + 14x)^2 - \operatorname{cosec}^2(e^x + 7x^2)\end{aligned}$$

- 3) We know that the rate of change, at time t_0 , in the dependent variable V is given by the value of the derivative $V'(t)$ at time t_0 .

$$V'(t) = 10t - 64$$

(a) As $V'(3) = 10(3) - 64 = -34 < 0$,

Therefore, the volume of water in the tank is decreasing

(b) As $V'(7) = 10(7) - 64 = 6 > 0$

Therefore, the volume of water in the tank is increasing

- (c) Here, the question is about only magnitude, irrespective of sign. At which point of time the magnitude is larger. The magnitude is larger at $t=3$.

$$(a) \quad dy = (3x^2 \sin 5x + 5x^3 \cos(5x)) dx$$

$$(b) \quad df = (-3e^{4-3w}) dw$$

$$(c) \quad dz = (\sin(y)e^{x \sin(y)}) dx + (x \cos(y)e^{x \sin(y)}) dy$$

$$(d) \quad dg = (5x^4 y^3 / z^2) dx + (3y^2 x^5 / z^2) dy + (-2 x^5 y^3 / z^3) dz$$

- 5) Calculate the directional derivative $D_{\vec{u}} f(\vec{x})$ for $f(x; y; z) = \sin(yz) + \log(x^2)$ at $(1, 1, \pi)$ in the direction of $\vec{v} = \langle 1, 1, -1 \rangle$

$$\nabla f(x, y, z) = \langle f_x, f_y, f_z \rangle = \langle 2/x, z \cos(yz), y \cos(yz) \rangle$$

$$\nabla f(1, 1, \pi) = \langle 2, \pi \cos(\pi), \cos(\pi) \rangle = \langle 2, -\pi, -1 \rangle$$

$$\|\vec{v}\| = \sqrt{(1)^2 + 1^2 + (-1)^2} = \sqrt{3} \neq 1$$

Hence, unit vector $\vec{u}$ in the direction of $\vec{v}$ is $\vec{u} = \langle 1/\sqrt{3}, 1/\sqrt{3}, -1/\sqrt{3} \rangle$

Thus,

$$\begin{aligned} D_{\vec{u}} f(x, y, z) &= \nabla f(1, 1, \pi) \cdot \vec{u} \\ &= \langle 2, -\pi, -1 \rangle \cdot \langle 1/\sqrt{3}, 1/\sqrt{3}, -1/\sqrt{3} \rangle \\ &= 1/\sqrt{3} \{2 - \pi + 1\} = (3 - \pi)/\sqrt{3}. \end{aligned}$$

6)

$$\text{Let } f_1(x, y) = x \log y, \quad f_2(x, y) = \sin x, \quad f_3(x, y) = x^2 y, \quad f_4(x, y) = y^3.$$

$$J_f = \begin{bmatrix} \partial f_1 / \partial x & \partial f_1 / \partial y \\ \partial f_2 / \partial x & \partial f_2 / \partial y \\ \partial f_3 / \partial x & \partial f_3 / \partial y \\ \partial f_4 / \partial x & \partial f_4 / \partial y \end{bmatrix} = \begin{bmatrix} \log y & x/y \\ \cos x & 0 \\ 2xy & x^2 \\ 0 & 3y^2 \end{bmatrix}$$

Therefore,

$$(J_f)_{(\pi/4, \pi/2)} = \begin{bmatrix} \log(\pi/2) & 2 \\ 1/\sqrt{2} & 0 \\ \pi^2/4 & \pi^2/16 \\ 0 & 3\pi^2/4 \end{bmatrix}$$

- 7) Let c be any arbitrary point of $[a, b]$, with $c \neq a$. We will show using mean value theorem that $f(c) = f(a)$. As c is an arbitrary point of $[a, b]$, value of f at every point of the interval equals to $f(a)$, implying f is constant on $[a, b]$.

As f satisfies all the conditions of Mean Value Theorem for interval $[a, b]$, it also satisfies the conditions for $[a, c]$, **because** $c \in [a, b]$. Therefore, by Mean Value Theorem, there exists d in the interval $[a, c]$ such that

$$(f(c) - f(a)) / (c - a) = f'(d) = 0 \quad (f'(x) = 0 \text{ for all } x, \text{ (given)}), \text{ implying}$$

$$f(c) - f(a) = 0, \text{ i. e. } f(c) = f(a) \text{ for all } c \text{ in } [a, b].$$

- 8) Let x_1 and x_2 be any two distinct points of $[a, b]$ such that $x_2 > x_1$. Then $[x_1, x_2] \subseteq [a, b]$ and f satisfies all the conditions of the Mean value theorem in $[x_1, x_2]$.

Therefore, there exists a number c with $x_1 < c < x_2$ and

$$(f(x_2) - f(x_1))/(x_2 - x_1) = f'(c)$$

Implying

$$(f(x_2) - f(x_1)) = f'(c)(x_2 - x_1) > 0 \text{ (as } x_2 > x_1, \text{ assumed; } f'(c) > 0 \text{ (given))}$$

Hence $f(x_2) > f(x_1)$, for $x_2 > x_1$; implying f is a strictly increasing function.

